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Methylimidazolium-Tetrafluoroborate Ionic Liquids Flowback Aids for Hydraulic Fracturing Applications

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Abstract

Hydraulic fracturing has been widely adopted to enhance hydrocarbon recovery in tight and ultra-low permeability reservoirs. However, a significant operational challenge persists due to the retention of fracturing fluids within the formation, resulting in reduced gas mobility and production efficiency. High capillary pressures, wettability alterations, and interfacial tension between fracturing fluids and formation fluids impede effective flowback, particularly under harsh reservoir conditions. The objective of this study was to evaluate the performance of tetrafluoroborate-based methylimidazolium ionic liquids (ILs) as flowback additives to reduce fluid retention, lower capillary pressure, and enhance fluid recovery under elevated temperature and salinity environments.

A series of structurally varied ILs were formulated by modifying the alkyl chain length while maintaining a common imidazolium cation and a tetrafluoroborate anion. These ILs were tested across multiple screening stages, including surface tension (ST), interfacial tension (IFT), and capillary pressure experiments. Seawater and formation brine were used as base fluids, while tight Kentucky sandstone cores with permeability around 0.1 mD were selected to simulate unconventional reservoir conditions. Surface tension and CMC determinations were performed using the Wilhelmy plate method, and thermal and salinity stability were evaluated using high-pressure, high-temperature pendant drop IFT measurements. Capillary pressure behavior was assessed using centrifuge experiments with short (3-hour) and long (20-hour) flowback cycles, complemented by NMR analysis to validate saturation trends.

The IL with the longest alkyl chain, ST4, demonstrated the strongest surface activity, reducing ST to ~27 mN/m and reaching CMC at ~300 mg/L. IFT values dropped from 70.56 mN/m to 22 mN/m in formation brine and from 65.48 mN/m to 27 mN/m in seawater at 120°C. In centrifuge tests, ST4 reduced residual water saturation (S_{wi}) from 36% to 30.5% and lowered entry capillary pressure by approximately 80%,

from 40 psi to 5 psi. NMR T_2 distributions further confirmed improved water removal and mobility in ST4-treated samples, highlighting its superior performance under extended flowback conditions.

These results confirm the robust thermal and salinity stability of ST4 and its effectiveness in modifying interfacial properties, lowering capillary pressure, and enhancing fluid displacement. The application of ILs, especially ST4, presents a novel and efficient approach to improving flowback performance in unconventional hydraulic fracturing operations.

Introduction

Hydraulic fracturing is a widely used well-stimulation technique aimed at enhancing hydrocarbon production by creating fractures in subsurface formations through the injection of high-pressure fluids. This method, introduced commercially in 1949 by Stanolind Oil & Gas Company (Clark, 1949), involves pumping a mixture of water, proppants, and chemical additives into the reservoir to generate and sustain fractures that improve the flow of oil and gas toward the wellbore. Over time, it has become a cornerstone of unconventional reservoir development, with over 2.5 million hydraulic fracturing operations performed globally by 2012 (Economides et al., 1994; Paktinat et al., 2011). Slickwater fluids, which are characterized by low viscosity and reduced friction, are now used in over 90% of fracturing treatments, particularly in shale and tight gas formations (Nasr-El-Din & Samuel, 2006; Al-Adwani & Sharma, 2002).

Despite their operational advantages, water-based fracturing fluids pose significant challenges in tight and ultra-low permeability reservoirs. A critical issue is water retention, where a large portion of the injected fluid remains trapped near the fracture faces and within the proppant pack. Studies have shown that only 10–30% of the injected fluid is typically recovered during flowback in gas wells (Holditch, 1979; Nasr-El-Din & Samuel, 2006). The retained water increases water saturation near the fracture face, resulting in a reduction in relative permeability to gas and a corresponding decrease in overall production efficiency. Additionally, this water retention results in both chemical and physical damage such as clay swelling and pore blockage (Zhang et al., 2015), and physically, it leads to water loading, capillary discontinuity, and flow restriction within the fracture system (Sharma & Agrawal, 2013; Abrams & Vinegar, 1985).

The capillary end effect is a particularly problematic phenomenon in tight formations, where capillary pressure differences between the matrix and fracture face trap water in the matrix. During production, hydrocarbons must displace this trapped fluid, a process that requires overcoming significant capillary resistance (Kianinejad et al., 2014; Rapoport & Leas, 1953). Liang et al. (2015) found that reducing matrix capillary pressure, rather than simply increasing the capillary number, could improve gas flow and enhance cleanup efficiency. Recovery of the retained fluid is further governed by interfacial tension and drawdown pressure, which together influence the effectiveness of capillary-driven cleanup (Paktinat et al., 2005; Rostami et al., 2014). Consequently, successful mitigation strategies must focus on reducing interfacial tension and improving drawdown potential to enable more effective hydrocarbon recovery in fractured low-permeability reservoirs.

Research on volatile fluids in fracturing treatments enhances water evaporation and improves gas permeability (Al-Anazi et al., 2005; Kamath & Laroche, 2003; Mahadevan & Sharma, 2005). This method aids in removing water from the wellbore and conductive fractures, but is limited in matrix regions with small production distances, making it inadequate to fully eliminate water blockage (Patzek et al., 2014). To tackle these challenges, flowback additives have been introduced to improve fluid recovery and mobilize trapped fluids. Additives like surfactants, microemulsions, nano-surfactants, and ionic liquids play vital roles in modifying interfacial properties, reducing capillary pressure, and enhancing flowback efficiency. The following sections will examine the evolution and effectiveness of these additives in addressing water retention issues and optimizing hydraulic fracturing operations.

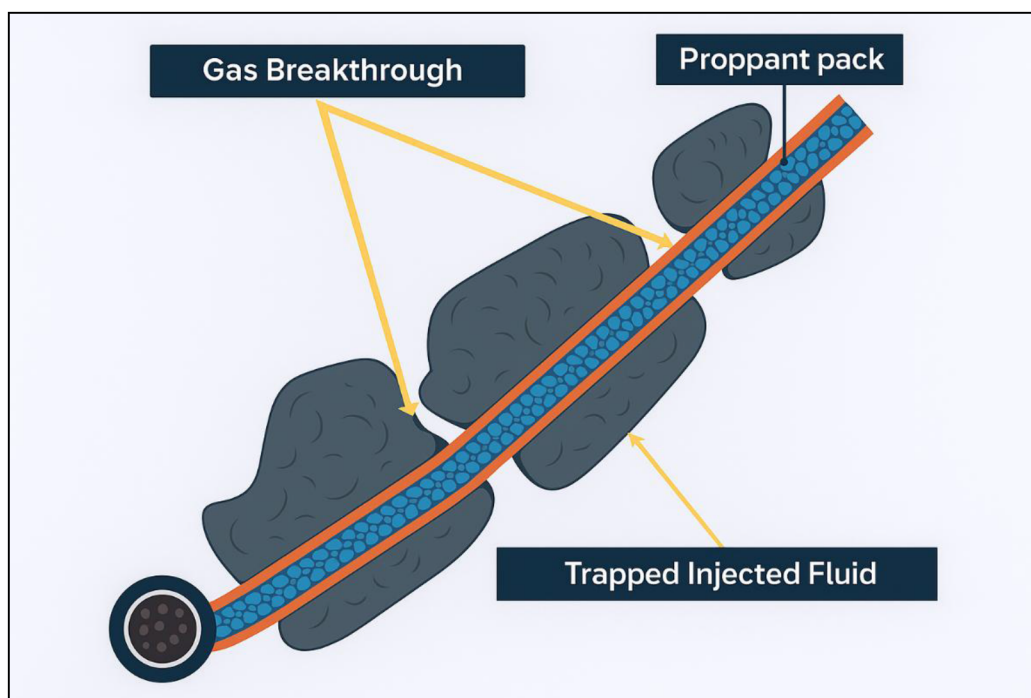


Figure 1—Water retention in gas reservoirs after Hydraulic Fracturing (after Pakitant et al., 2005)

Surfactants and microemulsions are widely used as flowback additives in hydraulic fracturing to enhance fluid recovery and improve hydrocarbon production. Surfactants function by reducing interfacial tension (IFT), altering wettability, and minimizing capillary forces that hinder hydrocarbon flow, especially in gas reservoirs where water retention near the fracture face can severely reduce gas permeability. Early research, such as by [Graham et al. \(1959\)](#) and [Poetket et al. \(1961\)](#), demonstrated the ability of surfactants to aid fluid recovery and improve acid stimulation. Various surfactant types—*anionic, cationic, zwitterionic, and non-ionic*—offer different performance profiles depending on reservoir conditions like salinity, clay content, and temperature. However, their effectiveness is often compromised by limitations such as adsorption onto rock surfaces (especially in clay-rich formations), thermal degradation, and salinity sensitivity, which reduce their ability to maintain low IFT and thus hinder their long-term performance in harsh reservoir environments ([Hamouda et al., 2006](#); [Wang et al., 1995](#); [Gupta et al., 2007](#)). To overcome some of these limitations, microemulsions have been introduced as more robust alternatives. Microemulsions are stable, transparent mixtures of water, oil, surfactants, and co-surfactants that exhibit ultralow IFT and improved penetration into low-permeability formations. Their thermodynamic stability and ability to solubilize both aqueous and oil phases make them highly effective in mobilizing trapped fluids and reducing water blocks. Studies by [Liang et al. \(2016\)](#) and [Rostami et al. \(2014\)](#) highlight their success in mitigating capillary end effects, improving gas flow, and enhancing permeability recovery. Microemulsions outperform conventional surfactants in several key metrics, including lower adsorption, higher stability under variable salinity and temperature conditions, and improved oil displacement. Research by [Mahmoudkhani et al. \(2015\)](#) and [Andrei et al. \(2009\)](#) demonstrates that microemulsions exhibit higher regain permeability and deeper penetration, even when surface tension and interfacial tension reductions are comparable to those achieved with surfactant-only treatments.

Despite their advantages, microemulsions are not without challenges. Their performance is highly dependent on formulation and salinity conditions. Experiments by [Liang et al. \(2016\)](#) demonstrated that different types of microemulsions—Type I (oil-in-water), Type II (water-in-oil), and Type III (bicontinuous)—exhibit variable flowback efficiencies and pressure drop behaviors under different salinity levels. Type III microemulsions, while stable, showed higher resistance to flow and increased fluid

retention. Additionally, salinity sensitivity remains a major limitation. Gupta et al. (2007) revealed that deviations from optimal salinity led to phase instability, increased IFT, and poor fluid displacement. For instance, surfactant NP showed a 2000-fold increase in IFT with just a slight increase in salinity, drastically reducing recovery efficiency. Thus, while surfactants and microemulsions provide essential mechanisms for improving flowback and hydrocarbon recovery, their application requires precise control over environmental conditions to ensure consistent and effective performance.

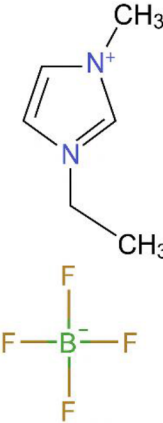
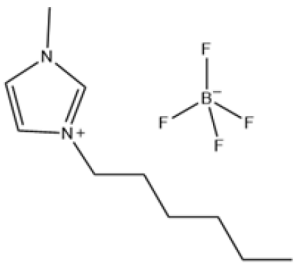
Ionic liquids (ILs) have emerged as promising alternatives to conventional surfactants for use as flowback additives in hydraulic fracturing, particularly in challenging unconventional reservoirs. Abdelgawad et al. (2023) demonstrated the potential of imidazolium-based ILs, such as 1-Decyl-3-methylimidazolium chloride, in reducing surface tension and interfacial tension (IFT), enhancing water-wetness, and improving permeability regain by 25%, with fluid recovery reaching up to 56%. ILs have distinct advantages over traditional surfactants due to their negligible vapor pressure, high thermal stability, tunable polarity, and compatibility with high-salinity and high-temperature environments. Structural variations in ILs, including cation types and alkyl chain lengths, provide tailored surface activity that can significantly lower IFT (often below 0.1 mN/m), enhance wettability alteration, and reduce capillary pressure near the fracture face. These properties make ILs well-suited for mitigating fluid retention and promoting efficient flowback in tight gas formations.

Among various IL families, imidazolium-based ILs are particularly effective due to their superior surface activity, thermal resilience, and viscosity tunability. Research has shown that longer alkyl chains on the imidazolium cation improve IFT reduction and lower surface tension, especially at elevated temperatures, which is critical for deep and hot reservoir environments. These ILs maintain moderate surface tension across a wide temperature range and can adapt viscosity for maintaining fracture integrity during treatment. Their thermal stability allows consistent performance in reservoirs exceeding 200°C, making them highly suitable for harsh fracturing conditions. Moreover, their tunable viscosity, low adsorption on rock surfaces, and ability to penetrate tight formations support their role as high-performance flowback aids. Overall, imidazolium-based ILs offer a robust, thermally stable, and efficient solution to overcome the limitations of traditional surfactants, positioning them as a next-generation additive for fluid recovery in unconventional reservoir development.

Material and Methodology

In this study, a series of methylimidazolium-based ionic liquids (ILs) were selected to evaluate their performance as flowback additives for hydraulic fracturing applications under challenging reservoir conditions. The ILs were systematically designed to vary in alkyl chain length while maintaining a standard imidazolium core structure, and all utilized tetrafluoroborate ($[\text{BF}_4^-]$) as the counter anion. This approach enables the assessment of how molecular structure, particularly alkyl chain length, affects surface tension, interfacial tension, and ultimately, fluid recovery. The four ILs investigated are 1-ethyl-3-methylimidazolium tetrafluoroborate, 1-butyl-3-methylimidazolium tetrafluoroborate, 1-hexyl-3-methylimidazolium tetrafluoroborate, and 1-decyl-3-methylimidazolium tetrafluoroborate, with increasing molecular weight and hydrophobicity from ST1 to ST4, respectively. Their chemical structures and molecular formulas are detailed in Table 1, illustrating the structural variation strategy used in this research.

Table 1—Chemical structure of the ILs for this study

1-Ethyl-3-methylimidazolium tetrafluoroborate	1-Butyl-3-methylimidazolium tetrafluoroborate	1-Hexyl-3-methylimidazolium tetrafluoroborate	1-Decyl-3-methylimidazolium tetrafluoroborate
$C_6H_{11}BF_4N_2$	$C_8H_{15}BF_4N_2$	$C_{10}H_{19}BF_4N_2$	$C_{14}H_{27}BF_4N_2$
			

To simulate reservoir-relevant environments, two distinct brine systems were utilized: synthetic seawater as the base fracturing fluid and native formation brine for Stability experiments. The base fluid has a total dissolved solids (TDS) concentration of 58,857 ppm, while the formation brine features significantly higher salinity, reaching 213,734 ppm. Table 2 outlines the ionic composition of both fluids, including key ions such as Na^+ , Cl^- , Ca^{2+} , and Mg^{2+} . These fluids were chosen to test the ILs under a wide range of salinity conditions, mirroring those commonly found in unconventional reservoirs. The ILs' ability to maintain performance in such highly saline environments is essential for evaluating their suitability as next-generation flowback additives.

Table 2—Ionic concentration for the formation brine and the fracturing base fluid

Ionic Species	Formation brine, ppm	Fracture base fluid, ppm
Na^+	59,491	18,414
K^+	0	577
Ca^{2+}	19,040	577
Mg^{2+}	2,439	2,040
Cl^-	132,060	32,031
SO_4^{2-}	350	5,206
HCO_3^-	354	--
TDS	213,734	58,857

The rock matrix used in this work consists of low-permeability Kentucky sandstone cores, selected to replicate the pore structure and mineral composition typical of unconventional formations. These cylindrical core plugs, shown in Table 3, were used for capillary pressure experiments. They feature a dominant quartz content (66%), along with significant amounts of plagioclase (17%), illite (14%), and minor K-feldspar (3%), representative of common tight gas and shale reservoirs. The permeability of these cores is

approximately 0.1 millidarcy (mD), making them ideal for assessing how ILs facilitate fluid recovery and reduce capillary-driven fluid retention in tight formations.

Table 03—The mineralogy of the tight Kentucky sandstone

Mineral	Quartz	Illite	K-Feldspar	Plagioclase
Wt%	66	14	3	17

Surface Tension Determination

In this study, the surface tension (ST) behavior of methylimidazolium-based ionic liquids (ILs) was evaluated using the KRÜSS K100 tensiometer, employing the Wilhelmy plate method. The primary goal was to assess each IL's ability to reduce the surface tension of the seawater-based fracturing fluid (TDS $\approx$ 57,000 ppm) and identify its Critical Micelle Concentration (CMC)—a key parameter indicating optimal surfactant performance. Surface tension plays a vital role in hydraulic fracturing, as lower surface tension (ST) values reduce capillary pressure and enhance fluid mobility, leading to improved flowback and cleanup efficiency. A series of ST measurements were conducted by incrementally increasing the IL concentration and recording the corresponding ST values.

As shown in Fig. 2, surface tension decreased steadily with increasing IL concentration due to the adsorption of surfactant molecules at the air–liquid interface. Beyond a specific concentration, the ST plateaued, signifying micelle formation in the bulk fluid. This inflection point in the curve represents the CMC and is clearly marked in the graph. The micelle formation is further visualized through schematic illustrations highlighting the transition from individual surfactant molecules to micellar aggregates. This graphical method enabled the precise determination of CMC values for each ionic liquid (IL), providing a foundational screening criterion. ILs with the most effective surface activity—based on their low CMC and substantial ST reduction—were selected for subsequent phases of the study, including thermal and salinity tolerance, and capillary pressure testing.

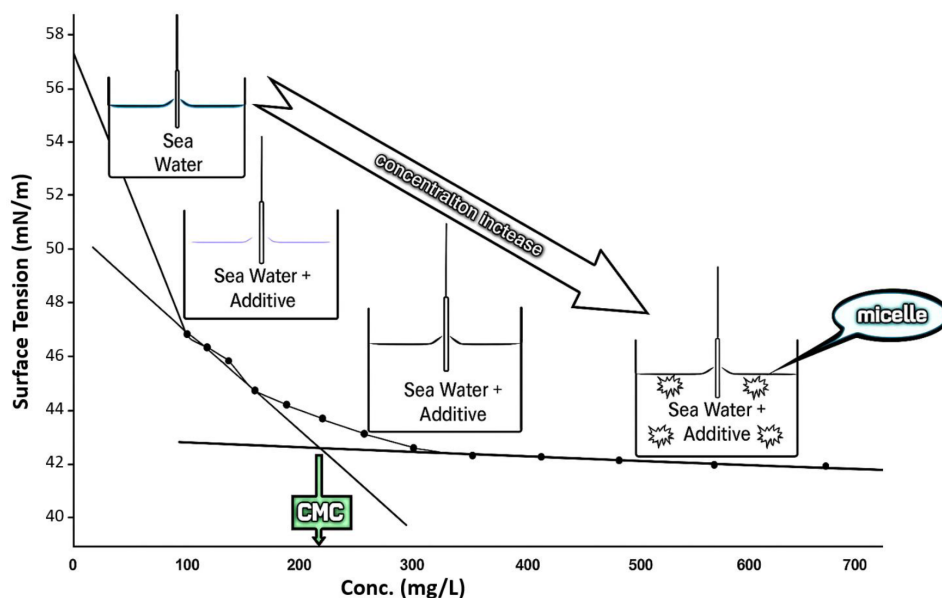


Figure 2—Surface tension vs. ionic liquid concentration, highlighting the Critical Micelle Concentration (CMC) for methylimidazolium-based ILs. The plateau indicates micelle formation.

Interfacial Tension Evaluation

To assess the performance of selected methylimidazolium-based ionic liquids (ILs) under harsh reservoir conditions, interfacial tension (IFT) measurements were conducted using the KRÜSS DSA 100 Drop Shape Analyzer, as shown in Fig. 3. This high-precision instrument enables accurate determination of IFT under elevated temperatures and pressures, simulating the subsurface environments typical of tight gas and shale reservoirs. The experiments were conducted at temperatures ranging from 30 °C to 120 °C and pressures up to 500 psi, with nitrogen gas serving as the hydrocarbon phase. The test fluids included salinity levels up to 21%, replicating formation brine composition. The pendant drop method was utilized, where a droplet is suspended in a nitrogen-filled chamber, and its shape, determined by the balance of gravitational and interfacial forces, is analyzed to calculate the interfacial tension (IFT).

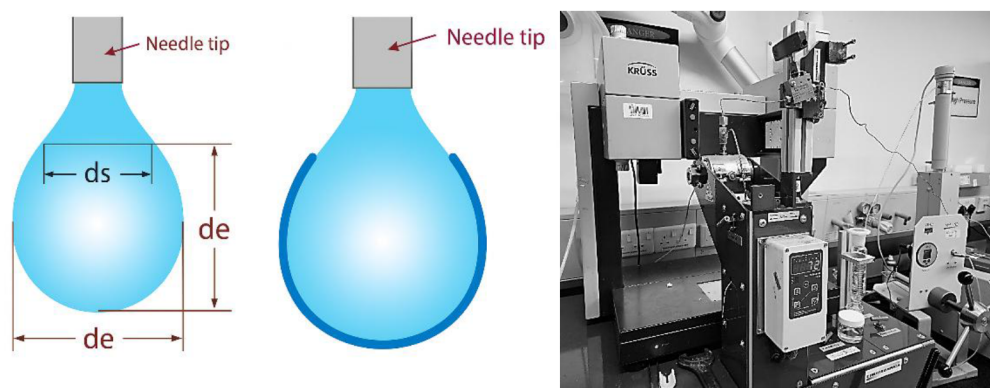


Figure 3—Schematic representation of the pendant drop method and KRÜSS DSA 100 Drop Shape Analyzer

The IFT values were derived from the droplet profile using the Young-Laplace equation, specifically focusing on the ratio of the droplet's equatorial and vertical diameters. This experimental approach was chosen to evaluate the thermal and salinity resilience of IL formulations, parameters where conventional surfactants often fail due to degradation, phase instability, or reduced effectiveness. As shown in the accompanying schematic of the pendant drop method and the experimental setup, this methodology provided critical insights into the robustness of ILs. The results supported the use of ionic liquids as durable and efficient flowback additives capable of maintaining interfacial performance under extreme reservoir conditions.

Capillary Pressure Measurements

Capillary pressure (CP) is a critical parameter that governs fluid retention and mobility in tight and ultra-low-permeability reservoirs. In such formations, high capillary forces often lead to the entrapment of fracturing fluids in nanopores, severely hindering hydrocarbon flow and reducing effective permeability. To mitigate this issue, flowback additives such as ionic liquids (ILs) are used to reduce capillary pressure, thereby enhancing fluid displacement and improving flowback efficiency. Following initial selection based on surface tension and interfacial tension (IFT) screening, the shortlisted ILs underwent further testing to evaluate their effectiveness in modifying capillary pressure behavior. As illustrated in Fig. 4, the experimental workflow involved determining porosity using a helium porosimeter, followed by core saturation via vacuum and pressure techniques to ensure full fluid saturation. Core weights before and after saturation were recorded to validate consistency with porosity data.

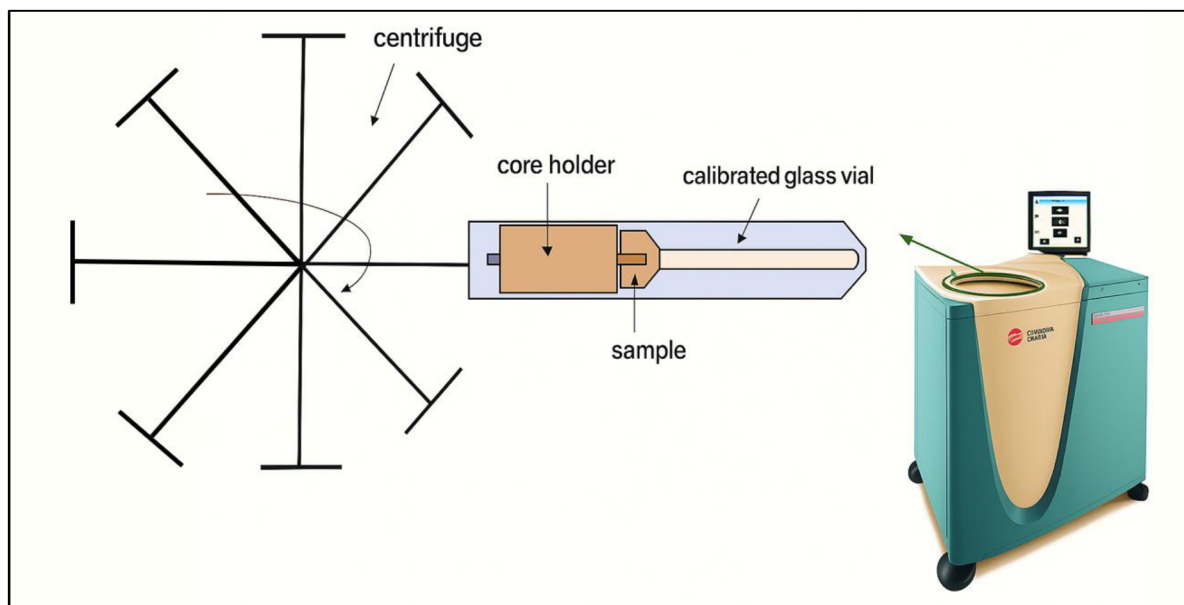


Figure 4—High-Pressure Centrifuge System utilized for accelerated capillary pressure experiments, allowing rapid evaluation of fluid recovery and entry pressure behavior.

To assess capillary pressure and fluid recovery performance, centrifuge-based testing was employed using a high-speed, high-pressure centrifuge system, depicted on the right side of Fig. 4. This setup enabled the simultaneous evaluation of three core samples, each mounted inside a calibrated glass vial within the centrifuge rotor. Two experimental runs were conducted to simulate different flowback durations: a short-term run with 3-hour spin periods at each rotational speed to mimic rapid flowback, and a long-term run with 20-hour spin durations to assess extended drainage effects. The centrifuge technique provided valuable insight into the impact of ILs on water retention, entry pressure, and overall fluid recovery. By comparing short and long flowback scenarios, the data helped determine the ILs' capability to improve recovery dynamics. The integration of centrifuge results with those from conventional porous plate methods ensured a more comprehensive evaluation of IL performance in real reservoir-like drainage conditions.

Results and Discussion

Results of Surface Tension and CMC Measurements

Figure 5 presents the surface tension behavior of four methylimidazolium-based ionic liquids (ILs)—ST1 to ST4—all containing the tetrafluoroborate anion, when added to a seawater-based fracturing fluid. As the concentration of ILs increases, a clear decline in surface tension is observed until a plateau is reached, signifying the Critical Micelle Concentration (CMC). This plateau indicates that micelle formation has begun, and further addition of IL does not result in significant surface tension reduction. Among the ILs tested, ST4 demonstrated the most notable reduction, lowering the surface tension to approximately 27 mN/m, while ST1, ST2, and ST3 stabilized at 50 mN/m, 47 mN/m, and 43 mN/m, respectively. This trend reflects significant differences in surface activity, with ST4 emerging as the most effective candidate for reducing surface tension.

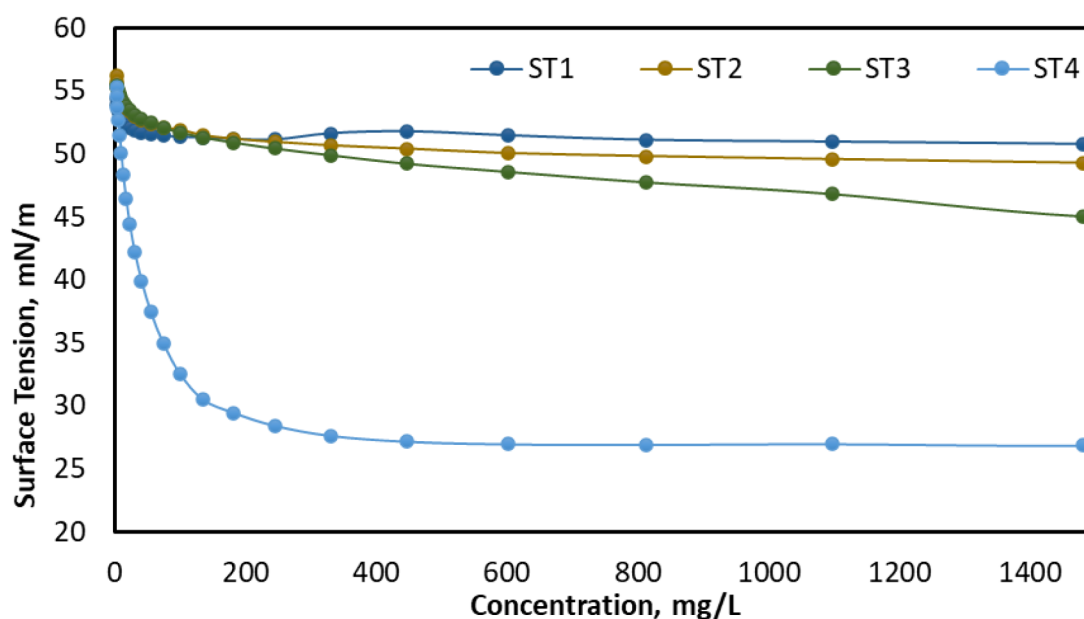


Figure 5—Surface tension as a function of ionic liquid concentrations for ST1, ST2, ST3, and ST4 in fracture base fluid.

The differences in performance across the ILs can be primarily attributed to the length of the alkyl chain attached to the imidazolium cation. ST1 contains an ethyl group, ST2 a butyl, ST3 a hexyl, and ST4 a decyl chain. Longer alkyl chains enhance the hydrophobicity of the molecule, which promotes more efficient molecular packing at the air–liquid interface and leads to stronger surfactant behavior. ST4’s ability to reach CMC at approximately 300 mg/L highlights its efficiency, requiring less material to achieve maximum surface activity. This translates to economic and operational benefits, as lower additive concentrations can be used while still achieving effective reduction in interfacial tension. The sharp surface tension drop seen with ST4 also indicates a high degree of molecular adsorption at the interface before micellization begins, a favorable trait in flowback additive design.

The performance of ST4 is further reinforced by comparisons with previous literature. [Tariq et al. \(2010\)](#) and [Jaeger & Eggers \(2009\)](#) reported similar findings, noting that ionic liquids with longer alkyl chains demonstrate sharper reductions in surface tension and improved surfactant-like behavior. The sharp decrease observed for ST4 in this study aligns closely with those results, validating that the molecular structure, particularly the extended hydrophobic tail, plays a decisive role in enhancing surface activity. This correlation not only supports the findings here but also underscores the relevance of molecular tailoring in optimizing IL formulations for specific engineering functions, such as hydraulic fracturing fluid flowback.

From an application standpoint, the results strongly suggest that ST4 is highly suitable for use as a flowback additive in hydraulic fracturing fluids, particularly in high-salinity systems such as seawater-based formulations. Its ability to reduce surface tension significantly at low concentration levels makes it ideal for improving fluid mobility in tight gas and shale formations, where capillary pressure can severely hinder flowback and hydrocarbon recovery. Furthermore, the enhanced performance of ST4 compared to other ILs tested demonstrates the importance of rational IL design, where the alkyl chain length is tuned to maximize interfacial efficiency. Future studies should explore how these molecular characteristics influence not only surface tension but also thermal stability, adsorption on rock surfaces, and overall recovery performance in dynamic coreflood environments.

Thermal and Salinity Stability Performance of ST4

The thermal stability of the ionic liquid ST4 was rigorously evaluated under conditions simulating harsh subsurface environments, with interfacial tension (IFT) measurements conducted at temperatures ranging

from 30°C to 120°C under a pressure of 500 psi. As shown in Fig. 6, these tests were performed across four fluid systems: formation brine, seawater-based fracturing fluid (SW), and their respective counterparts treated with 0.5 gpt of ST4. In both fluid systems, ST4 significantly reduced IFT values across all temperatures. For instance, in the formation brine, which initially exhibited high IFT due to elevated salinity, the addition of ST4 decreased IFT from 70.56 mN/m at 30°C to just 22 mN/m at 120°C. A similar trend was observed in seawater, with IFT dropping from 65.48 mN/m to 27 mN/m when ST4 was added. These results confirm that ST4 retains strong surface activity even under extreme thermal and salinity conditions.

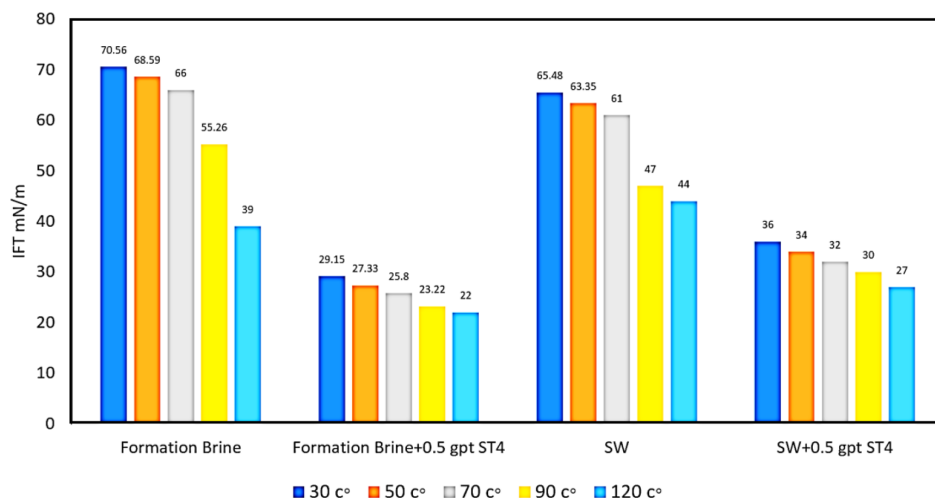


Figure 6—Interfacial tension (IFT) between nitrogen and various fluids, with and without ST4, was measured across different temperatures.

The pendant drop method, visualized in Fig. 7, further supports these findings. As the temperature increased, the droplet shape progressively elongated and narrowed, indicating reduced interfacial tension between nitrogen gas and the liquid phases. This visual confirmation correlates with the measured IFT values and suggests that ST4 enhances interfacial mobility at elevated temperatures by improving molecular alignment at the gas-liquid interface. Notably, no visual signs of instability, phase separation, or surfactant degradation were detected up to 120°C, which is often beyond the operating range of conventional surfactants. This highlights ST4's thermal resilience and its suitability for high-temperature fracturing operations.



Figure 7—Pendant drop images showing temperature-driven droplet shape changes for brine with ST4 and untreated seawater.

In seawater systems, where the salinity is moderate, the IFT reductions achieved with ST4 were also substantial, albeit slightly less dramatic than those observed in formation brine. The lower baseline salinity of seawater results in less interfacial resistance initially, but the presence of ST4 still produced a significant reduction in IFT across all temperature levels. More interestingly, the enhanced performance in formation brine suggests that high concentrations of divalent cations such as Ca^{2+} and Mg^{2+} may promote stronger adsorption of ST4 at the interface or facilitate earlier micelle formation, further aiding IFT reduction. This ion-assisted behavior could be a unique advantage of ST4, reinforcing its effectiveness in highly saline environments common in unconventional plays.

Overall, the results demonstrate that ST4 exhibits exceptional thermal and salinity stability, maintaining performance across a wide operational envelope. The ability to sustain low IFT values under elevated temperature and brine salinity makes ST4 a robust and versatile flowback additive. Its compatibility with both moderate (seawater) and high-salinity (formation brine) fluids, alongside its consistent behavior observed via both quantitative IFT data and qualitative drop shape analysis, positions ST4 as a high-performing candidate for flowback enhancement in tight and shale gas reservoirs where conventional additives often fall short.

Capillary Pressure Results and Interpretation

Capillary pressure (P_c) experiments were conducted using a high-speed centrifuge system to evaluate the impact of ST4—an ionic liquid with a tetrafluoroborate anion—on water displacement and flowback performance under varying flowback durations. These tests simulate conditions in tight formations where capillary forces significantly hinder fluid recovery. Two distinct flowback periods were examined: a short duration of 3 hours per rotational step and a long duration of 20 hours. Prior to testing, all core samples were fully characterized for porosity and permeability using helium porosimeter and gravimetric saturation methods to ensure accurate capillary pressure interpretation. As shown in Fig. 8, during the long flowback period (20 hours), ST4 consistently outperformed the untreated base fluid. The capillary pressure curves reveal that ST4-treated cores exhibited lower P_c values across all water saturations, confirming its enhanced ability to mobilize retained water. Most notably, residual water saturation (S_{wi}) decreased from 36% for the base fluid to 30.5% for ST4, indicating more effective drainage and reduced capillary trapping. The extended flowback duration amplified these effects by allowing more time for the IL to act on the interface, lowering resistance to fluid flow. A closer examination of the high-saturation region ($S_w > 90\%$) reveals a particularly significant trend. At 95% water saturation, the base fluid required a P_c of approximately 40 psi to initiate flowback, whereas the ST4-treated sample required only 5 psi. This represents an 80% reduction in entry pressure, a dramatic improvement that highlights ST4's superior ability to weaken capillary barriers and facilitate fluid movement under conditions where conventional fluids typically struggle. This performance is crucial in unconventional reservoirs, where entry pressures can significantly impact the success of initial cleanup and hydrocarbon flow initiation.

In summary, the capillary pressure tests confirm that ST4 significantly enhances flowback performance by reducing both residual water saturation and entry capillary pressure. The longer flowback duration further reinforces the additive's effectiveness by allowing sustained interaction with the rock-fluid interface. These results validate ST4's potential as a robust and efficient flowback aid for tight and ultra-low-permeability reservoirs, where mitigating capillary forces is crucial for enhanced production efficiency.

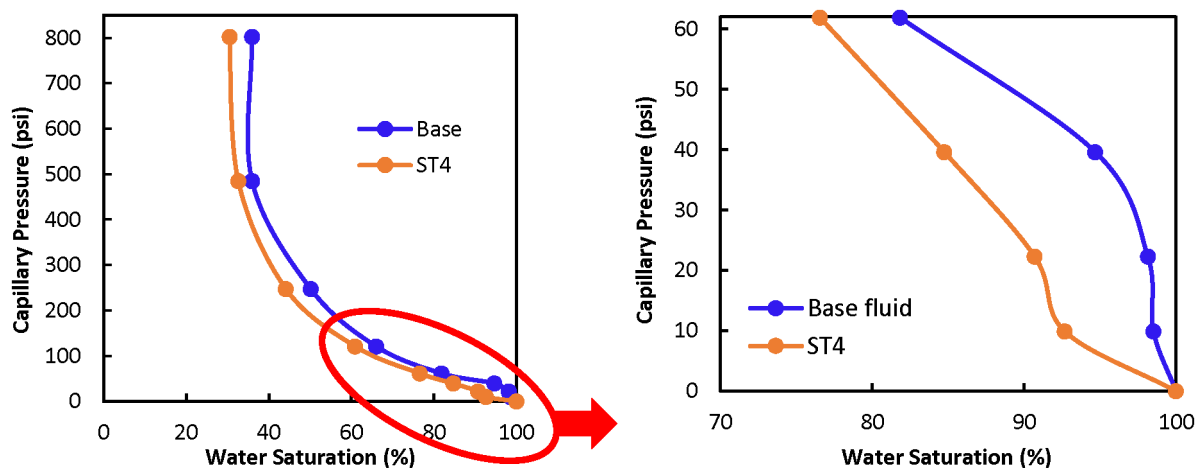


Figure 8—Capillary pressure (P_c) vs. water saturation (S_w) curves comparing short and long flowback periods. Insets highlight the high-saturation range ($S_w > 90\%$), where ST4 reduced entry pressure at $S_w = 95\%$ by $\sim 80\%$, demonstrating enhanced water mobilization during extended flowback.

The set of centrifuge experiments, conducted over a 20-hour flowback period, provided critical insight into the efficiency of ionic liquids in reducing capillary pressures. ST4 significantly outperformed the base fluid, lowering the entry capillary pressure from approximately 40 psi to just 5 psi—an 80% reduction. This dramatic drop confirms ST4’s enhanced ability to mobilize trapped water in tight formations where capillary forces typically inhibit flowback. The improved displacement performance underscores ST4’s potential in enhancing cleanup and early production in unconventional gas reservoirs.

To validate these findings, Nuclear Magnetic Resonance (NMR) measurements were conducted to analyze T_2 relaxation time distributions of the core samples, both before and after capillary pressure testing. As shown in Fig 9, the ST4-treated samples retained significantly less bound water after flowback, as indicated by the reduced signal in shorter relaxation times and a shift toward longer T_2 values. These shifts correspond to improved water removal and more effective pore drainage. The NMR cumulative porosity plots further confirm that ST4 led to greater fluid displacement compared to the base fluid, supporting the centrifuge-derived reductions in residual water saturation (S_{wi}).

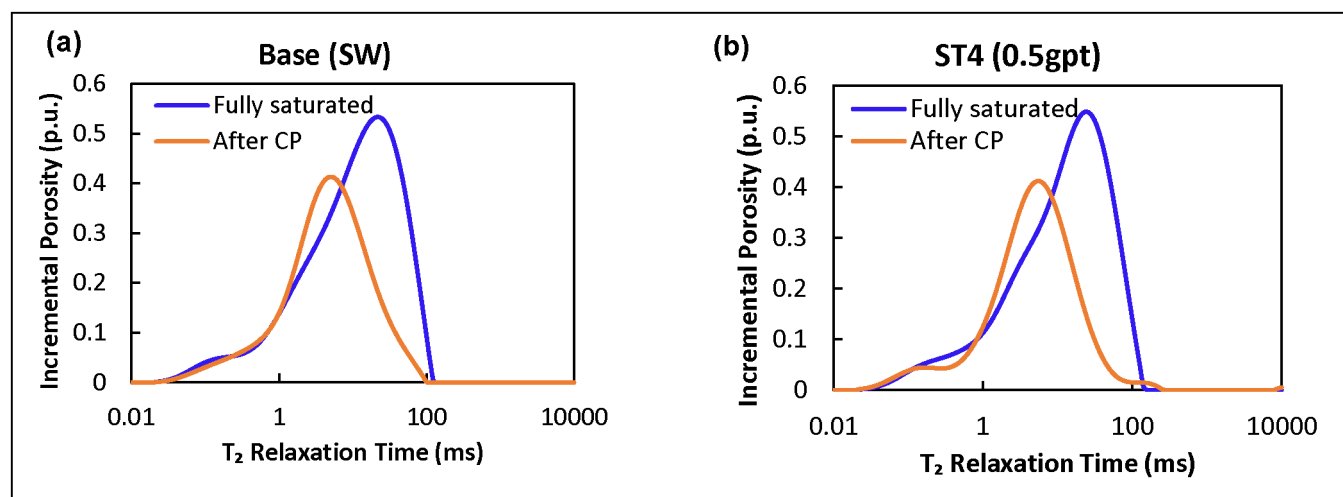


Figure 9— T_2 relaxation time distributions and cumulative porosity curves for cores treated with base fluid (a) and ST4 (b) after long flowback period, illustrating improved fluid mobility and reduced water retention in ST4-treated samples (c).

The comparison of incremental porosity curves also highlights the difference in saturation profiles between the two fluids. The base fluid retains more signal intensity in small pores (shorter T_2 times),

indicating higher water trapping, while ST4-treated samples show broader distributions and higher mobility, characteristic of better recovery. These outcomes validate the role of ST4 not only in reducing entry pressures but also in enhancing bulk fluid mobility through wettability alteration and interfacial tension reduction—mechanisms critical for efficient flowback in tight sandstone formations.

Overall, the combination of centrifuge and NMR data provides strong evidence that ST4 significantly improves water recovery by lowering capillary barriers and enhancing pore-level drainage. Its performance under long-duration flowback conditions confirms its robustness and effectiveness as a next-generation flowback additive for hydraulic fracturing, making it well-suited for deployment in high-salinity, low-permeability reservoirs.

Conclusions

This study investigated the performance of tetrafluoroborate-based methylimidazolium ionic liquids (ILs) as flowback additives in hydraulic fracturing applications, focusing on their ability to reduce interfacial tension (IFT), lower capillary pressure, and enhance water recovery under high-temperature, high-salinity conditions. A systematic evaluation was conducted using surface tension, interfacial tension, capillary pressure, and NMR-based fluid mobility analyses to assess the effectiveness of ionic liquids (ILs) in tight sandstone reservoirs. These are the main findings:

- The IL with the longest alkyl chain, ST4, achieved the lowest surface tension value of ~ 27 mN/m and reached the critical micelle concentration (CMC) at approximately 300 mg/L, indicating strong surface activity with low dosage requirements.
- In high-salinity formation brine at 120°C, IFT was reduced from 70.56 mN/m to 22 mN/m when treated with 0.5 gpt of ST4, while in seawater, the IFT dropped from 65.48 mN/m to 27 mN/m.
- Capillary pressure experiments showed that ST4 reduced residual water saturation (S_{wi}) from 36% to 30.5% under long-duration flowback conditions, confirming improved drainage efficiency.
- The entry pressure was lowered by approximately 80%, from 40 psi (base fluid) to 5 psi (ST4-treated), demonstrating its capability to overcome capillary barriers.
- NMR analysis validated these improvements, with ST4-treated cores showing better water displacement and reduced fluid retention.
- The thermal stability of ST4 was maintained up to 120°C, with no degradation observed, confirming its applicability in high-temperature fracturing environments.

These findings highlight the novelty of using structurally tailored ionic liquids (ILs), particularly ST4, as high-performance, thermally stable flowback additives that enhance cleanup efficiency and reduce water retention in unconventional hydraulic fracturing operations.

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